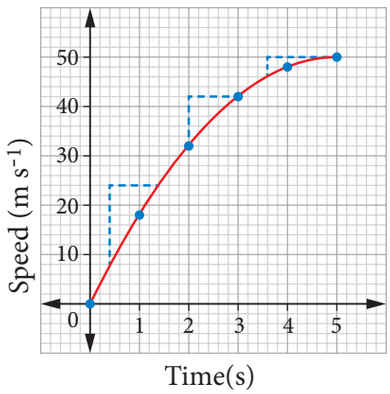
f. Bus travelling with decreasing acceleration (non-uniform acceleration)

Time (s)	0	1	2	3	4	5
Speed (ms^{-1})	10	18	32	42	48	50



The speed is decreasing with time. Hence, the graph has a positive and decreasing gradient, and the acceleration decreases.

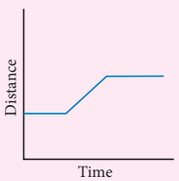
2.5.1 Comparison between Distance – Time and Speed – Time Graphs

The Speed – Time graphs and Distance – Time graphs may look very similar. But, they give different information. We can differentiate them by looking at the labels.

Graph and Story

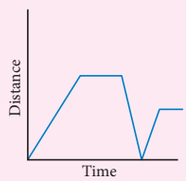


Raju began walking to his school. Suddenly he remembered that he forgot his pen and walked back home. But he stopped suddenly when he heard a noise.



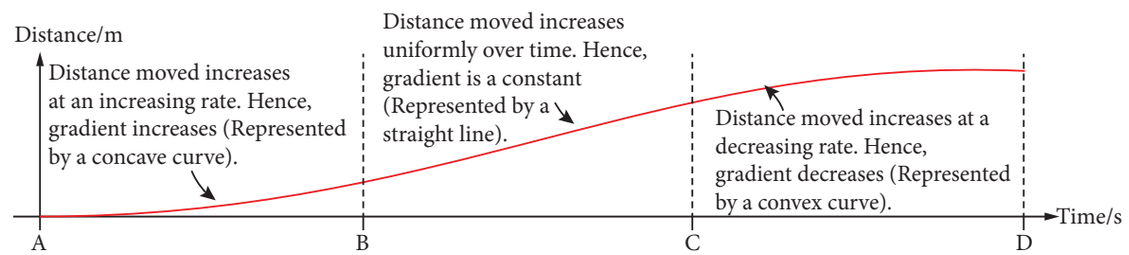
Rani was waiting for her mother for some time. When she saw her mother, she ran out of her home hugged her and stood there for a while.

Imagine and write a story on your own for the given graph?

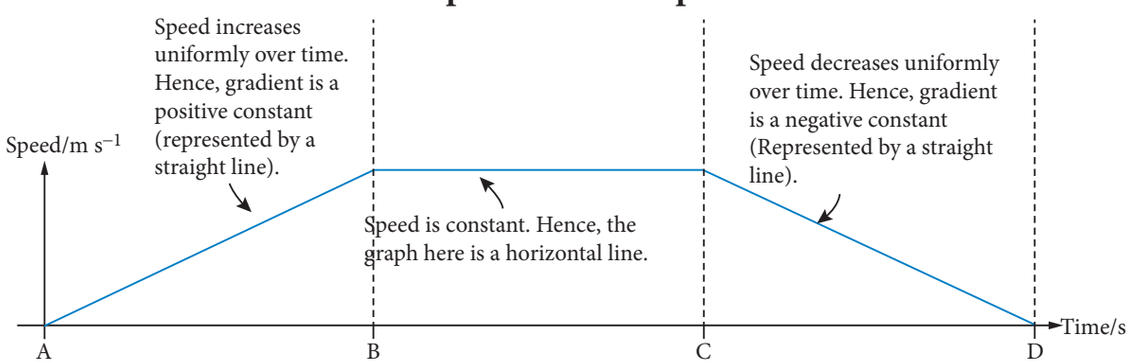


From A to B	From B to C	From C to D
Car accelerates uniformly from rest.	Car moves at constant speed.	Car decelerates uniformly to a stop.

Distance–Time Graph



Speed–Time Graph



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